

Report for research position UGent/KUL

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Documentation of my thought process. An honest tale of time constraints and overambitious approaches

I want to begin with a general note on the approach that I wanted to give to this challenge. After checking what the provided dataset contained and which modeling requests you had, it was clear that the end goal was to create a predictive model with a binary output: reject/approve a loan. Immediately, I thought that most candidates were going to provide a logistic regression/ max likelihood framework or a simple Machine-learning model. Discontent with the prospect of mimicking what most applicants could trivially think of, I had begun to rush through a novel approach, for which I took inspiration from a course that I took during the last semester here at KUL (Dynamic Methods). I proceeded to attempt to model the lender's decision to approve or reject a loan as a dynamic discrete choice problem similarly to Rust's 1987 model of bus engine replacement. Some of the similarities were the existence of a binary choice by an agent; forward looking behavior (expected profit/ losses) with expected (binary) utility functions; state variables that can be modeled to output conditional choice probabilities (in the original model the engine mileage, here a vector of borrowers' characteristics). Notwithstanding these similarities, the absence of i) a time dimension to render the model dynamic and ii) data on rejected applications rendered my initial efforts futile, as I inexorably realized that workarounds would result in a clunky/overkill model with no predictive-accuracy augmenting capabilities compared to more elementary alternatives. I hence resorted to modeling in a more basic manner as you rightfully suggested from the beginning. I did keep one aspect of the Rust model in my logistic regression setup, and will display more information on the first approach in the final notes of this report nonetheless, so as to give you an idea of what I originally had in mind. I should lastly mention that I attempted a quick xgboost model once the data was loaded and the predictive difference was minimal (below $\frac{1}{100}$) with my original logit, so I stuck with the first one as I am more familiar with the interpretation.

1 Data

Overall, the provided dataset is surprisingly comprehensive in terms of potential predictive information. For the analysis I decide to include all the variables but "customer id", "application date", "zip code prefix", "payment to income ratio". Besides the first two which are not relevant in a logit model, the reason as to why I drop the zip code has to do with THE EU AI act (also overfitting). I realize that some of the covariates that I use are likely correlated (some are by construction linearly dependent like loan amount and loan amount to income) but by running a Variance Inflation Factor I ended up dropping only payment to income ratio for a high VIF. Regardless, the accuracy of the model doesn't seem to depend on slightly tweaked specifications. Because of this (and for the sake of time), I also refrain from dropping other regressors that might display multicollinearity (i.e., income and education). As for more general data tweaking, I converted repaid/defaulted loans in Boolean and constructed dummy variables where needed (i.e. education, instead of having it as years of schooling). When it comes to potentially missing data, if there was a possibility to include profits and losses coming from a loan (assuming no standardization of rates), then data on interest rates (from which profits and losses can be derived) is missing. Further, information about rejections and time repeated observations for individuals could be valuable if dynamic models similar to the sketch that I propose could be fundamental.

2 Set up

Model Specification and Estimation

Let $y_i \in \{0, 1\}$ denote a binary indicator for loan default for borrower i , and let $x_i \in \mathbb{R}^K$ represent the vector of explanatory variables, consisting of standardized continuous features and one-hot encoded categorical dummies as also reported in the code.

I estimate a binary logistic regression model:

$$\Pr(y_i = 1 \mid x_i) = \Lambda(x_i^\top \beta) = \frac{1}{1 + \exp(-x_i^\top \beta)}$$

The parameter vector β is estimated via maximum likelihood by solving:

$$\hat{\beta} = \arg \max_{\beta} \sum_{i=1}^n [y_i \log \Lambda(x_i^\top \beta) + (1 - y_i) \log (1 - \Lambda(x_i^\top \beta))]$$

Here is the part that I kept from the thought experiment of the DDCM mentioned earlier. Instead of utilizing the standard probability threshold of logit to predict outcomes, I introduce the possibility to move this threshold on the basis of the expected return or loss of the loan. Even though these values are not in the dataset, possible idiosyncratic tweaking might help the decision of loan approvals. Given the predicted probability $\hat{p}_i = \Lambda(x_i^\top \hat{\beta})$, I define the expected profit from approving the loan as:

$$\pi_i = (1 - \hat{p}_i) \cdot R + \hat{p}_i \cdot (-L) = R - \hat{p}_i(R + L)$$

The loan is approved if $\pi_i > 0$, which is equivalent to the threshold rule:

$$\text{Approve loan if } \hat{p}_i < \frac{R}{R + L}$$

Note that if $R = L$, the threshold collapses to a standard logit.

3 Output

The overall accuracy of the prediction, estimated as the vaules where predicted(default)=actual(default) and predicted(repaid)=actual(repaid) over the total sample was 76.65 percent. I also plotted the best predictor of repayment against the probability of defaults (not the binary actual outcome) and found that a 100-point increase in credit score reduces the odds of default by a staggering 77 percent circa, ceteris paribus. Some of these findings are very intuitive, for instance, when loans are for cars and home improvements (usually less lavish sums), the repayment becomes more likely. Others are less so (application type), but I will not dwell on interpretations and data description, as I am positive that your expertise far exceeds my rushed reporting intuition.

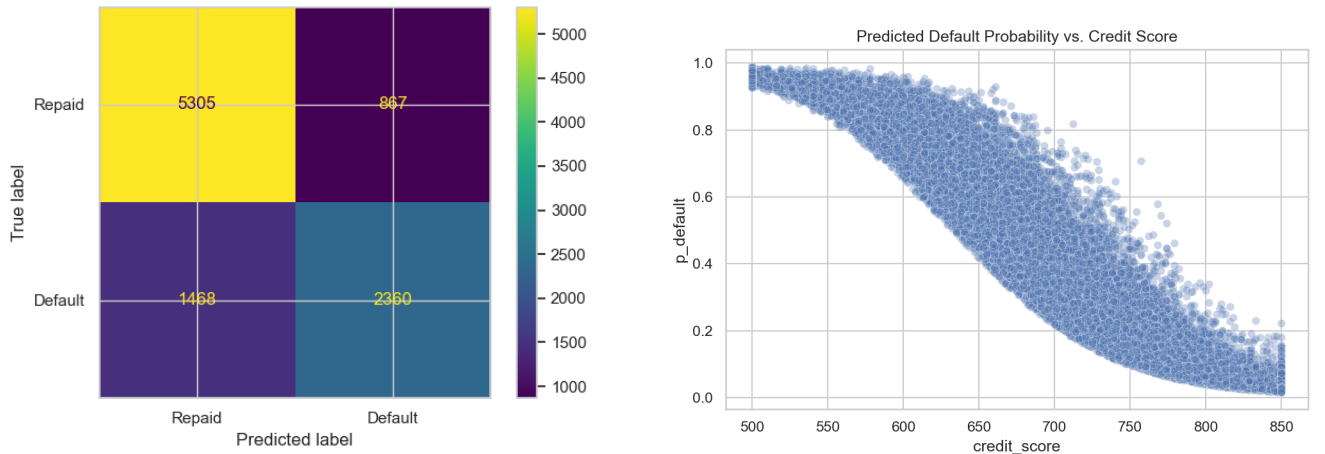


Table 1: Top Predictors of Default and Repayment (Logistic Regression Coefficients)

Default Predictors	Coefficient	Repayment Predictors	Coefficient
Debt to income ratio	0.306305	Employment length years	-0.133705
Previous defaults	0.262226	Savings amount	-0.140551
Loan to income ratio	0.186079	Loan purpose: car	-0.176402
Application type: online	0.142070	Loan purpose: home improvement	-0.199011
Application type: phone	0.133804	Credit score	-1.494007

In the code, I also include a measure of adjusted McFadden’s pseudo R^2 for the model, with a value of 0.2665, but I would be cautious with a possible interpretation for the sake of assessing predictive performance.

4 EU AI Act brief discussion

I will premise this section by noting that I found some of the points mentioned on the github page rather nebulous (i.e., what is meant by risks, who are the users etc.). Going through the most relevant points (enumerated as in the original page):

- 1 If risks are related to the data management system, this goes beyond the ex-post analysis of data. If by risk it is implied errors in credit scoring or loan approvals, once again having repeated observations of individuals (i.e., when certain variables change) can potentially improve the accuracy of such scores.
- 2 The data was given pre-cleaned for this challenge, but it is of importance to ensure error minimization to have a system that detects potential inaccuracies (missing values, divisions by 0 etc)
- 3,4 and 5 Besides mantining a log of changes in the model (i.e., due to changes in policy), the point on transparency is fundamental: information about credit scoring/ loan approvals should be both easy to read to applicants and legal supervisors and have a technical documentation as to how it operates. High levels of transparency foster non-discrimination practices and inform individuals who might be critical about its implementation.
- 6 Human ovverride has to be ensured for peculiar situations, though it should always be warranted to prevent potential corruption/ conflicts of interest
- 7 Security against data manipulation is not only in the interest of applicants, but in that of the bank. For this reason, it should be robust to data leakages/ reverse engineering on certain data (i.e., exactly deducing income/wealth from other data). More complex models with more parameters should provide a better solution in this regard.
- 8 As mentioned in the introduction, I believe that the zip code could be intentionally excluded, as it often captures socioeconomic status, local infrastructure, or even race/ethnicity.

If the end-goal of Prometislab is that of maximizing the profit of a bank by helping in developing an optimal loan approval rule,

5 EXTRA: Sketch of the dynamic discrete choice model a la Rust

State Variables

Let $s \in \mathcal{S}$ represent the observable state of a borrower, including features such as credit score, income, and loan amount.

Decision Variable

The lender chooses $d_t \in \{0, 1\}$:

$$d_t = \begin{cases} 1 & \text{if the loan is approved} \\ 0 & \text{if the loan is rejected} \end{cases}$$

Per-Period Utility Function

The per-period utility is defined as:

$$u(s, d) = \begin{cases} \pi(s) + \varepsilon_1 & \text{if } d = 1 \quad (\text{approve}) \\ 0 + \varepsilon_0 & \text{if } d = 0 \quad (\text{reject}) \end{cases}$$

where:

$$\pi(s) = p(s) \cdot R - (1 - p(s)) \cdot L$$

is the expected profit from approving the loan, with:

- $p(s)$: probability of repayment given state s
- R : reward/profit from repayment
- L : loss incurred from default

The shocks $\varepsilon_0, \varepsilon_1$ are i.i.d. Extreme Value Type I.

Bellman Equation

The value function satisfies:

$$V(s) = \max_{d \in \{0, 1\}} \{u(s, d) + \beta \mathbb{E}[V(s') \mid s, d]\}$$

Using the logit closed form, I get:

$$V(s) = \log(\exp(\pi(s) + \beta \mathbb{E}[V(s') \mid s, d = 1]) + \exp(\beta \mathbb{E}[V(s') \mid s, d = 0])) + \gamma$$

where γ is Euler's constant.

Choice Probability

The lender's choice probability of approval is:

$$P(d = 1 \mid s) = \frac{\exp(\pi(s) + \beta \mathbb{E}[V(s') \mid s, d = 1])}{\exp(\pi(s) + \beta \mathbb{E}[V(s') \mid s, d = 1]) + \exp(\beta \mathbb{E}[V(s') \mid s, d = 0])}$$

Given access to longitudinal data on borrower states, including both approved and rejected applications as well as observed state transitions over time, it would be possible to formulate a fully dynamic discrete choice model. This would allow for value function iteration, estimation of transition probabilities $P(s' \mid s, d)$, and the derivation of optimal approval policies. Moreover, such a model could be used to simulate counterfactual scenarios, such as the effect of macroeconomic shocks on default risk.